Day 15
15 of 30 · Sundays holiday

# Laser physics & laser safety
లేజర్ ఫిజిక్స్ & సేఫ్టీ

TODAY'S OBJECTIVES · ఈ రోజు లక్ష్యాలు

- Explain wavelength, fluence, pulse width and spot size simply
- Follow laser room safety like a professional
- Prepare, check and shut down a laser machine correctly

THEORY · థియరీ

## Four dials

Wavelength (nm) decides what the light is absorbed by and how deep it goes — 755 alexandrite, 808 diode (hair), 1064 Nd:YAG (deeper, safer in dark skin), 1320/1064 picosecond (pigment). Fluence (J/cm$^2$) is energy per area — the power. Pulse width (ms) is how long the energy is delivered — longer is safer in dark skin. Spot size decides depth and speed.

## Selective photothermolysis

Choose a wavelength absorbed by the target (melanin in hair, pigment in a spot) more than by surroundings, and a pulse shorter than the target's cooling time. The target heats and is destroyed; the skin around it stays safe. In dark skin the epidermis also holds melanin — so we reduce fluence, lengthen pulse and cool aggressively.

## Laser room safety

Wavelength-specific goggles for client, operator and everyone in the room. Warning sign and closed door, no reflective jewellery, no flammable products (alcohol) on the skin, smoke evacuator for ablative work, machine on standby whenever not firing, foot pedal only under your own foot, test shot on a discreet area, documented settings for every session.

## Hands-on today

- Identify the machines in the clinic: wavelength, indications, contraindications
- Set up the laser room to safety standard and be audited
- Do a test shot on a volunteer's discreet area with the trainer and observe reaction at 10 minutes

## Safety & contraindications

- Never fire without correct goggles on everyone
- Never treat over tattoos, moles or recently tanned skin
- Know where the emergency stop is on every machine

### KEY POINTS TO REMEMBER · గుర్తుంచుకోవలసినవి

- 1064 Nd:YAG safest in dark skin
- Longer pulse = safer
- Test shot always
- Goggles for everyone in the room

## Homework

Write the safety checklist for your laser room, 12 points, and memorise it.

## Grooming & professional tip

The operator who does a test shot every time never has a bad week. Shortcuts in laser are not speed — they are risk.

### SELF-CHECK · సెల్ఫ్ చెక్

- Did I follow hygiene and consent for every client today?
- Which step did I need help with — and what will I drill tonight?
- Did I write my records and settings before leaving?

Trainer sign: _____ Trainee sign: _____

### MY NOTES · నా నోట్స్